

Captive-Habitat FTPL in an NK model

Session summary — numerical implementation, the calibration trilemma, and what it means

L. Leturia · prepared as a working note · code in `nk/habitat/` (full detail in `REPORT.md`)

Executive summary

We built the **microfounded** two-agent preferred-habitat FTPL model (the spec in *habitat_nk_spec.pdf*) that replaces the Entrega III reduced-form mandate: a short-habitat agent A and a long-habitat agent B optimise portfolios, a regulatory mandate forces B onto long gilts, A absorbs the residual, and the price wedge and fiscal rent *emerge* from this distortion. The steady state, log-linearisation, Klein/QZ solution and impulse responses all run and pass the spec's internal checks.

The headline scientific result is a **calibration trilemma**: the microfoundation **cannot simultaneously** deliver (i) a plausible habitat-preference strength, (ii) a data-consistent debt maturity structure, and (iii) the measured 14% wedge. The reduced form hid this by conflating two different 'long shares'. Surfacing the trilemma is the sharp contribution of going micro.

The trilemma in one table

Both columns use the same headline configuration (see p.2); the **only** difference is how the long-bond share $\bar{\chi}$ is interpreted. Both solve, pass all steady-state checks, and are determinate.

	Cal A: $\bar{\chi}$ read as a quantity share	Cal B: $\bar{\chi}$ set to the data market value
Quantity long-share $\bar{\chi}$	0.353	0.023
Market-value long-share w_L	0.93	0.35 (= DMO data)
Habitat strength ϕ (to hit $\bar{w}=14\%$)	3.0 — plausible	37.9 — implausibly large
Fiscal rent $\bar{X}$ / debt	0.34%	0.11%
λ +1% tightening → long price	+0.15%	+0.72%
λ +1% tightening → inflation	+0.47%	+1.20%
Monetary 25bp → output	-25.2%	-8.7%

Read it diagonally. Cal A buys a believable habitat penalty ($\phi \approx 3$) but the debt is then 93% long by market value, which makes the price level hyper-sensitive (a 25bp rate move implies a 25% output swing). Cal B fixes the maturity to the data, which cuts that hyper-sensitivity roughly threefold — but the wedge now needs a habitat penalty an order of magnitude larger. You can move along this frontier; you cannot get off it.

One-line version for the meeting

“A 14% market-value wedge generated from a realistically small long-bond captive position requires an implausibly strong habitat preference; making the preference plausible instead forces a counterfactually long-dominated debt stock and a hyper-elastic price level. That trade-off is the result.”

What the model is, and how it solves

Two optimising agents (population split 50/50) hold one short and one long nominal bond subject to a quadratic 'habitat' penalty for deviating from a preferred long share θ . The regulator's mandate binds on B's long position and replaces B's long-bond Euler equation; A prices both bonds, so the government is priced by A's stochastic discount factor. The block sits inside a standard NK environment (Calvo Phillips curve, passive Taylor rule). We compute the distorted, binding-mandate steady state numerically, log-linearise (20 equations), and solve with the Entrega III Klein/QZ routine.

Internal checks the implementation passes

- All six of the spec's steady-state sanity checks (interior allocations, positive wedge, mandate binds with margin, correct determinant sign, ...).
- The -1 inflation-revaluation coefficient (an identity that must hold exactly).
- Blanchard–Kahn determinacy: 9 stable / 11 unstable roots, with the relative-wealth 'closing device' root sitting just inside the unit circle at $1-\epsilon$, as required.
- A non-trivial cross-check: the fiscal rent computed from the cross-sectional portfolio formula equals what the government valuation identity independently requires, to machine precision.

The key modelling decisions we made

Decision A — the mandate redistributes, it does not expand supply. In the first pass the aggregate long supply expanded one-for-one with the mandate, so agent A was never actually squeezed and the long price *fell* on a tightening (wrong sign). Switching to a non-accommodating issuance rule — the mandate redistributes a fixed long stock from A to B — squeezes A off habitat, widens the wedge, and makes the long price *rise* on a tightening, as the financial-repression story requires.

Vayanos–Vila term premium (calibrated to gilt data). Agent A is the arbitrageur; it demands a duration-risk premium that captive demand *compresses*. We calibrated its strength to UK gilt data (Bloomberg pull): a $\sim 1\%$ /yr term premium at the model's duration gives $\theta_{tp} \approx 0.05$. It more than doubles the long-price response on a tightening (good for the mechanism) but is far too small to cure the hyper-elasticity.

Realistic shock sizes. The early 'insane' numbers were partly a unit-shock artifact (a unit innovation here is a $\sim 100\%$ / thousands-of-bp shock). Responses are now reported per 1% regulatory tightening and per 25bp monetary move.

Two things that do NOT fix the hyper-elastic price level

- **Stickier prices — wrong lever.** Inflation here is pinned by the fiscal valuation identity, not by the Phillips curve; the Phillips curve only backs out the output needed to deliver that inflation ($\hat{y} \approx \hat{\pi} / \kappa$). Making prices stickier (smaller κ) leaves inflation unchanged and makes output *explode*. We verified this.
- **A Bohn-style surplus-feedback / s-shaped surplus — helps the path, not the impact.** Cochrane's partial-repayment device smooths inflation over time and avoids the deficits-cause-deflation puzzle, but the impact jump is set by the valuation identity, so it barely moves on impact. The only lever that touches the monetary impact is reducing long-debt dominance — i.e. the calibration choice above.

A clarification you flagged — Leeper (1991) and 'partial Ricardian'

The **regime** is genuinely either/or: a unique equilibrium needs exactly one active and one passive authority (active money + passive fiscal, or passive money + active fiscal). You cannot blend regimes. But **'passive fiscal' is a threshold, not a point**. A surplus rule that responds to debt with coefficient γ is still *active* fiscal (still FTPL, still determinate under passive money) for any γ below the debt-stabilising threshold. So a positive-but-below-threshold debt response — 'partial Ricardian' — changes magnitudes without changing the regime. And Cochrane's s-shaped surplus is a *different axis entirely*: it shapes the time-series of surplus innovations (deficit-then-repayment), which is orthogonal to the active/passive classification, so it never touches the Leeper threshold. Both of these are consistent with your reading; they live *inside* the active-fiscal region, not across the regime boundary.

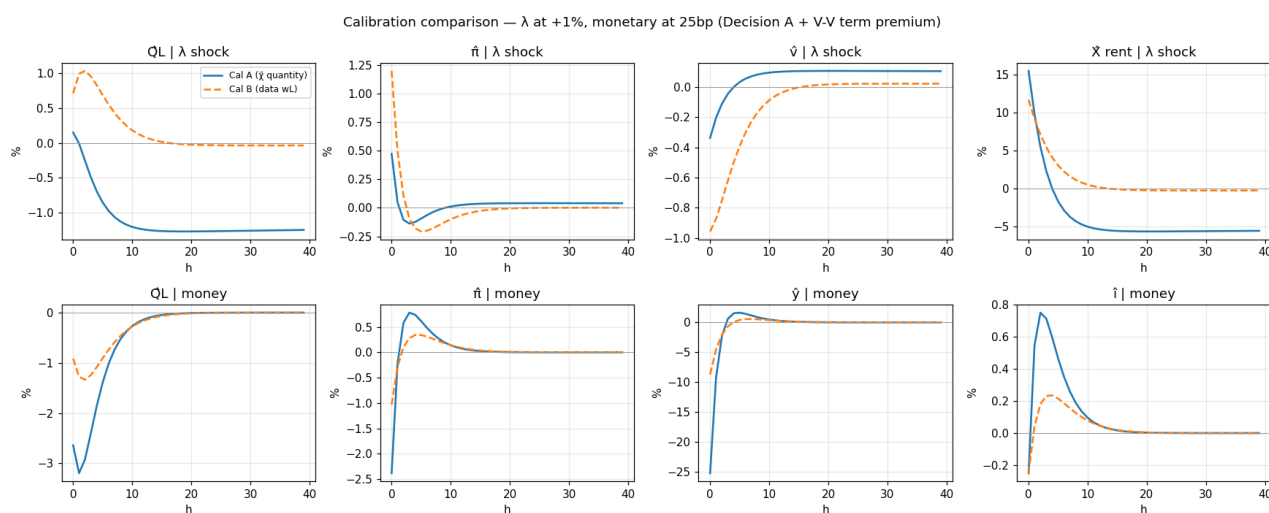
Why the signs and magnitudes come out as they do

Two of the spec's predicted signs are conditional and come out 'against' the reduced form, and the price level is hyper-elastic. Both trace to a single fact: with a long perpetuity (decay $\delta=0.96$, duration $\sim 5y$, priced at ~ 23), even a modest quantity share is a dominant *market-value* share, so the long-bond revaluation channel governs the price level (the FTPL-with-long-term-debt regime of Cochrane and Corhay–Kung–Morales). In that regime a contractionary monetary shock is **disinflationary** — this is documented in the literature and can be kept and cited. And because the fiscal rent is small relative to the debt (0.1–0.3%), the valuation identity ties the long price and inflation together: you cannot get 'long price rises' *and* 'inflation falls' at the same time unless the rent is a materially larger share of backing.

What I'd raise with the advisor

- **The calibration choice is yours to make and it is the headline.** Report Cal B (data-consistent dynamics, honest about the large ϕ) as the main case, Cal A as the 'what makes the penalty plausible' comparison — and let the gap between them be the result. Or accept the trilemma framing explicitly.
- **Can the rent be made first-order?** A larger captive sector μ_B (the rent scales with it) might relieve the trilemma by raising $\bar{X}$ /debt so the rent genuinely backs the price level — worth a quick exploration.
- **Duration vs the captive bucket.** The model's δ implies a 5-year bond, but the LDI/insurance captive demand is for 15–30y gilts. Matching the long end deepens the long-debt dominance — the same tension as the trilemma.
- **Deferred (your call):** a genuinely risk-based, time-varying term premium needs a higher-order / Epstein–Zin solution; we used a reduced-form stand-in for now.

Impulse responses under the two calibrations (λ at +1%, monetary at 25bp)



Solid = Cal A ($\bar{X}$ as quantity); dashed = Cal B (data market value). Note how Cal B (dashed) damps the monetary output and inflation responses.

Code & artifacts: [nk/habitat/](#) — [params.py](#), [steady_state.py](#), [linearize.py](#), [klein.py](#), [run_habitat.py](#), [calibrate.py](#), [sweeps.py](#), [compare_calibrations.py](#); full write-up in [REPORT.md](#); figures in [figures/](#).